Customer Life Cycle Value and Personalization with GenAI Advisor

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Problem Statement

Banks can't identify high-value customers, struggle and send generic offers, and have no predictive intelligence

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Business impact: lost revenue, low engagement, inefficient marketing spend

Solution

1. CLV prediction engine

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2. customer segmentation

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3. personalization engine

4. GenAI RAG chatbot

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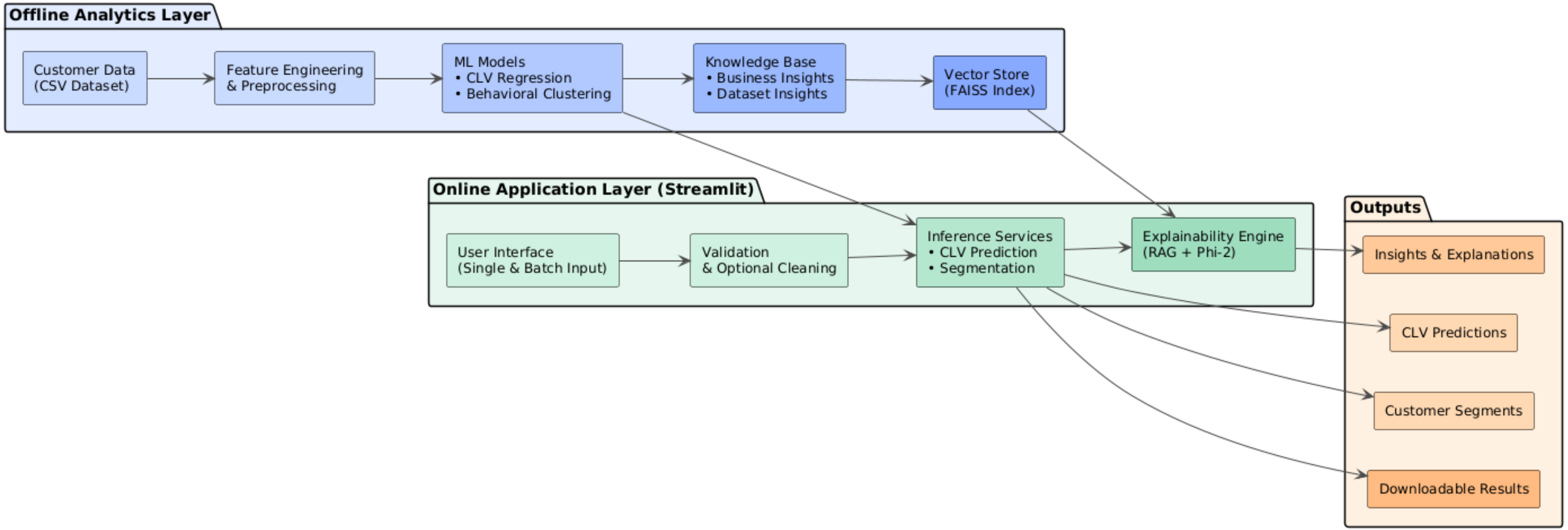
5. Streamlit dashboard

SYSTEM ARCHITECTURE



Technology Stack

Core Prrogramming language- Python

CSV Files- Input

Pandas, Numpy- data preprocessing and feature engineering

RFM Analysis- recency, frequency, monetary features

Scikit learn-CLV Regression model, Kmeans clustering, Silhouette analysis, Feature scaling

Phi-2(local model) -Text embeddings, RAG answer generation

FAISS- vector similarity search and retrieval

RAG pipeline- chunking amd retrieval augmented generation

Streamlit- web UI

Data

- Synthetic dataset of 100,000 customer records created for CLV model training
- Each row represents a single customer with transaction and behavioral details
- Key inputs include frequency, spend, recency, tenure, and average monthly spending

CLV prediction

- Developed a CLV regression model using engineered RFM and behavioral features

	R^2	Adjusted R2	RMSE
XG Boost	0.9945	99.45%	$73
Random Forest	0.9942	99.41%	$75
Linear Regression	0.6772	67.70%	$735

Customer Segmentation

- Used K-Means clustering to discover behavior-based customer segments
- Grouped customers based on purchasing patterns and value-related features
- Trained a Decision Tree to translate cluster behavior into interpretable segmentation rules
- Final output categorized customers into Low, Medium, and High value segments

K-MEANS EVALUATION METRICS

RAG Chatbot

OPUS

- Developed a **Retrieval-Augmented Generation (RAG) chatbot** to **explain and interpret CLV predictions**, without performing CLV computation itself.

- Created a **knowledge base from business logic and dataset-driven customer behavioral insights** generated during the CLV modeling process.

- Embedded knowledge chunks using **Phi-2** and stored them in a **FAISS vector index** to enable fast semantic retrieval.

- For each user query, the chatbot **retrieves relevant contextual information** from FAISS and uses **Phi-2 to generate grounded, explainable responses**

Challenges faced

-High latency due to local Phi 2 model
-Chunking Strategy tradeoffs
-Lack of real world noisy data

Key Learnings

-Clear business definitions are critical
-Data design influences learning depth
-Chunking is not just a preprocessing step

Future Enhancements

-Optimising RAG performance
-Advanced explainability
-Real world data integration and logging
-Larger Phi family models

Thank You

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